

# WEEK-4:

## Output:

NOTE: DON'T FORGET TO SAVE AND BACK UP YOUR COMPLETED JUPYTER NOTEBOOK AND LAB LOGBOOK, AND UPLOAD THE LAB LOGBOOK REQUIREMENTS AS A SCREENSHOT OF YOUR CODE AND RESULT(S) ON YOUR GITHUB.

```
[28]: # Import libraries
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
from sklearn.datasets import fetch_california_housing
SID = 2463439
first_hidden = int(str(SID)[-3:])
second_hidden = first_hidden // 2
print("Hidden layer sizes:", first_hidden, second_hidden)
data = fetch_california_housing()
X = data.data
y = data.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = keras.Sequential([
    layers.Input(shape=(X_train.shape[1],)),
    layers.Dense(first_hidden, activation='relu'),
    layers.Dense(second_hidden, activation='relu'),
    Dense(1)
])
model.compile(
    optimizer=Adam(),
    loss='mae',
    metrics=[MAE()]
)
```

verbose=1

```
)
test_loss, test_mae = model.evaluate(X_test, y_test, verbose=0)
print(f"\n✅ Test MAE: {test_mae:.4f}")
```

Hidden layer sizes: 439 219

Model: "sequential\_10"

| Layer (type)     | Output Shape | Param # |
|------------------|--------------|---------|
| dense_25 (Dense) | (None, 439)  | 3,951   |
| dense_26 (Dense) | (None, 219)  | 96,360  |
| dense_27 (Dense) | (None, 1)    | 220     |

Total params: 100,531 (392.70 KB)

Trainable params: 100,531 (392.70 KB)

Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
413/413 — 5s 6ms/step — loss: 11.4733 — mean_absolute_error: 11.4733 — val_loss: 12.7313 — val_mean_absolute_error: 12.7313
Epoch 2/10
413/413 — 2s 5ms/step — loss: 5.0897 — mean_absolute_error: 5.0897 — val_loss: 4.9923 — val_mean_absolute_error: 4.9923
Epoch 3/10
413/413 — 2s 6ms/step — loss: 3.5584 — mean_absolute_error: 3.5584 — val_loss: 2.5962 — val_mean_absolute_error: 2.5962
Epoch 4/10
413/413 — 2s 5ms/step — loss: 1.9394 — mean_absolute_error: 1.9394 — val_loss: 2.6205 — val_mean_absolute_error: 2.6205
Epoch 5/10
413/413 — 2s 5ms/step — loss: 1.9423 — mean_absolute_error: 1.9423 — val_loss: 0.7623 — val_mean_absolute_error: 0.7623
Epoch 6/10
413/413 — 2s 5ms/step — loss: 1.4315 — mean_absolute_error: 1.4315 — val_loss: 1.9417 — val_mean_absolute_error: 1.9417
Epoch 7/10
413/413 — 2s 5ms/step — loss: 1.2944 — mean_absolute_error: 1.2944 — val_loss: 0.8284 — val_mean_absolute_error: 0.8284
Epoch 8/10
413/413 — 2s 5ms/step — loss: 1.0691 — mean_absolute_error: 1.0691 — val_loss: 0.6419 — val_mean_absolute_error: 0.6419
Epoch 9/10
413/413 — 2s 5ms/step — loss: 1.0124 — mean_absolute_error: 1.0124 — val_loss: 1.2647 — val_mean_absolute_error: 1.2647
Epoch 10/10
413/413 — 2s 5ms/step — loss: 0.8495 — mean_absolute_error: 0.8495 — val_loss: 1.6462 — val_mean_absolute_error: 1.6462
```

✅ Test MAE: 1.6241

```
[29]: print(f"Mean Absloute Error in test data: {test_mae:.4f}")
```

Mean Absloute Error in test data: 1.6241

[ ]:

## Program:

```
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    layers.Input(shape=(X_train.shape[1],)),
    layers.Dense(first_hidden, activation='relu'),
    layers.Dense(second_hidden, activation='relu'),
    Dense(1)
])
model.compile(
    optimizer=Adam(),
    loss='mae',
    metrics=[MAE()]
)
model.summary()
history = model.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=10,
    batch_size=32,

    verbose=1
)
test_loss, test_mae = model.evaluate(X_test, y_test, verbose=0)
print(f"\n✅ Test MAE: {test_mae:.4f}")
```